

WOLKITE UNIVERSITY
DEPARTMENT OF MANAGEMANT
OPERATIONS RESEARCH
Model examination

Choose the best answer from the given alternative

1. The first step in solving Operations Research problem is:
A. Model building
B. **Formulation of the problem**
C. Obtain alternate solutions
D. Obtain basic feasible solutions
2. The constraints of Maximization problem are of:
A. Greater than or equal type
B. **Less than or equal type**
C. Less than type
D. Greater than type
3. The slack variables indicate:
A. Excess resource available
B. Shortage of resource available
C. Nil resources
D. **Idle resource**
4. To convert $\leq$ type of inequality into equations, we have to:
A. Assume them to be equations
B. Add surplus variables
C. Subtract slack variables
D. **Add slack variables**
5. To convert $\geq$ type of inequality into equations, we have to:
A. Add slack variable
B. Subtract slack Variable
C. **Subtract surplus variable**
D. Add surplus variable
6. The key row indicates:
A. Incoming variable
B. **outgoing variable**
C. Slack variable
D. Surplus variable
7. The key column indicates:
A. Outgoing variable
B. **Incoming variable**
C. Independent variable
D. Dependent variable
8. In solving maximization LP models using simplex algorithm, the iteration is terminated when
A. **$C_j - Z_j \leq 0$**
B. $C_j - Z_j = 0$
C. $C_j - Z_j \geq 0$
D. $Z_j \leq 0$
9. The solution of the Linear programming problem in graphical solution lies in:
A. First quadrant
B. **Second quadrant**
C. Third quadrant
D. Fourth quadrant
10. The number at the intersection of key row and key column is known as:
A. Column number
B. Row number
C. **Key number/Element**
D. Cross number

11. If an artificial variable exists in the final simplex tableau, the type of solution is
- Unbounded
 - Non optimal
 - Infeasible**
 - Degeneracy
12. _____ deals with the distribution of goods from several points of supplies (sources) to a number of points of demands (destinations).
- transportation problem**
 - assignment problem
 - simplex approach
 - Dual leaner problem
13. Which one is odd approach?
- north west corner method (NWCM)
 - least cost method (LCM)
 - Vogel's approximation method (VAM)
 - Hungarian Method**
14. Among type of decision making criterion _____ is very optimistic.
- Maximin
 - Minimax
 - Maximax**
 - Laplace
15. In Northwest corner method the allocations are made:
- Starting from the left hand side top corner**
 - Starting from the right hand side top corner
 - Starting from the lowest cost cell
 - Starting from the lowest requirement and satisfying first
16. The total number of allocation in a basic feasible solution of transportation problem of m - rows and n -column size is equal to:
- $m \times n$
 - $(m / n) - 1$
 - $m + n + 1$
 - $m + n - 1$**
17. In Hungarian method of solving assignment problem, the row opportunity cost matrix is obtained by:
- Dividing each row by the elements of the row above it
 - Subtracting the elements of the row from the elements of the row above it
 - Subtracting the smallest element from all other elements of the row
 - Subtracting all the elements of the row from the highest element in the matrix**
18. The decision making criterion follow selecting the alternative that has the best (maximum) of the worst payoffs is called.
- Maximin**
 - Minimax
 - Maximax
 - Regret
19. The beginning and end points of an activity are called;

- A. Events
- B. Nodes
- C. Dangling
- D. A & B

20. Activities that cannot be started until one or more of other activities are completed are called:

- A. Successor activities
- B. Predecessor activities
- C. Concurrent activities
- D. None

21. In transportation problem the solution is optimal when all of the unoccupied or empty cell evaluation result is

- A. Positive or/and zero
- B. Negative
- C. positive and negative
- D. zero or/and negative

22. _____ method of evaluating a transportation solution for optimality involves the use of index numbers that are established for the rows and columns.

- A. Modi method
- B. stepping stone method
- C. Least cost method
- D. VAM

23. Which one is not considered as

Additional variable in solving linear programming problem especially simplex method

- A. Slack variable(S)
- B. Artificial variable(A)
- C. Basic variable
- D. Surplus variable(-S)

24. Operations Research came into existence during:

- A. World War I
- B. India and Pakistan War
- C. World War II
- D. Stone Age

25. _____ is the variable that has the most positive value in the $C_j - Z_j$ row in simplex method

- A. leaving variable
- B. Entering variable
- C. Iterative variable
- D. Non basic variable